LAB6 – Troubleshooting Network Services (DNS, HTTP, FTP)

Course: The Bits and Bytes of Computer Networking
Author: Florencia Horita
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1. Introduction

The objective of this laboratory is to identify and solve failures in different network services within a local network using Cisco Packet Tracer software.

The DNS, HTTP, and FTP services were configured on the same server. Unlike the previous laboratory, in this case intentional failures were introduced in each service in order to apply troubleshooting techniques and analyze network behavior using Simulation Mode.

This approach makes it possible to understand how to detect network problems, identify their cause, and apply the appropriate solution.

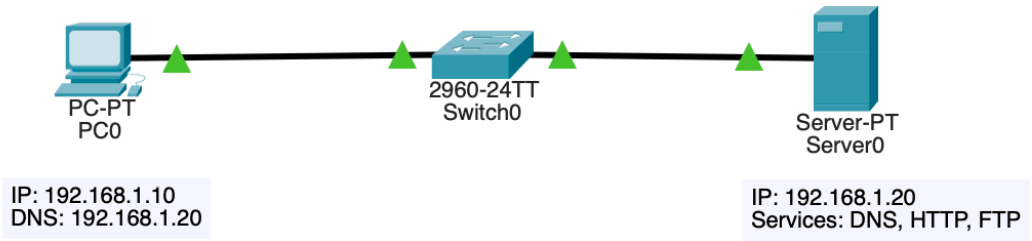
2. Network Topology

The network used in this laboratory consists of:

- One PC acting as a client
- One Cisco 2960 switch
- One server providing network services (DNS, HTTP, and FTP)

All devices are connected within the same local area network (LAN) through the switch, allowing communication between the client and the server.

Figure 1 – Network Topology



3. Network Configuration

Static IP addresses were configured to allow communication between devices.

Device	IP Address	DNS
PC0	192.168.1.10	192.168.1.20
Server0	192.168.1.20	—

The server was configured to provide DNS, HTTP, and FTP services, allowing the client to access the server using the domain **www.test.local**.

Figure 2 – PC IP Configuration

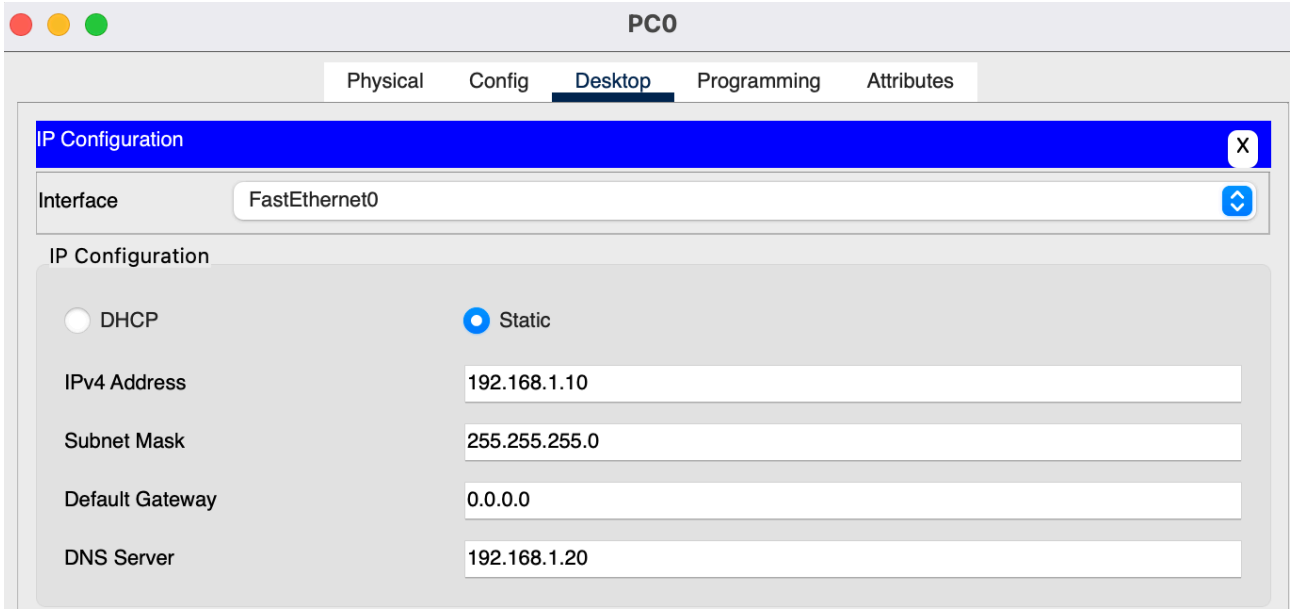
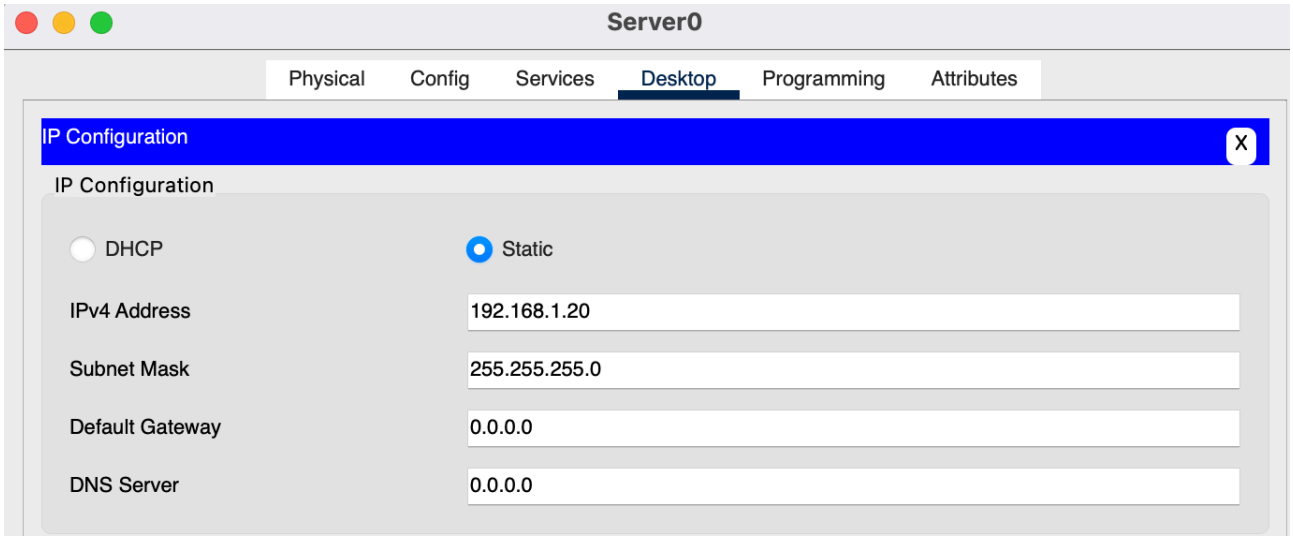


Figure 3 – Server IP Configuration



4. Initial Tests (Baseline)

Before introducing failures, all services were verified to be working correctly. DNS resolution, web page access via HTTP, and FTP server connection were tested.

These initial tests establish a baseline to compare network behavior when failures occur.

Figure 4 – DNS Service Working

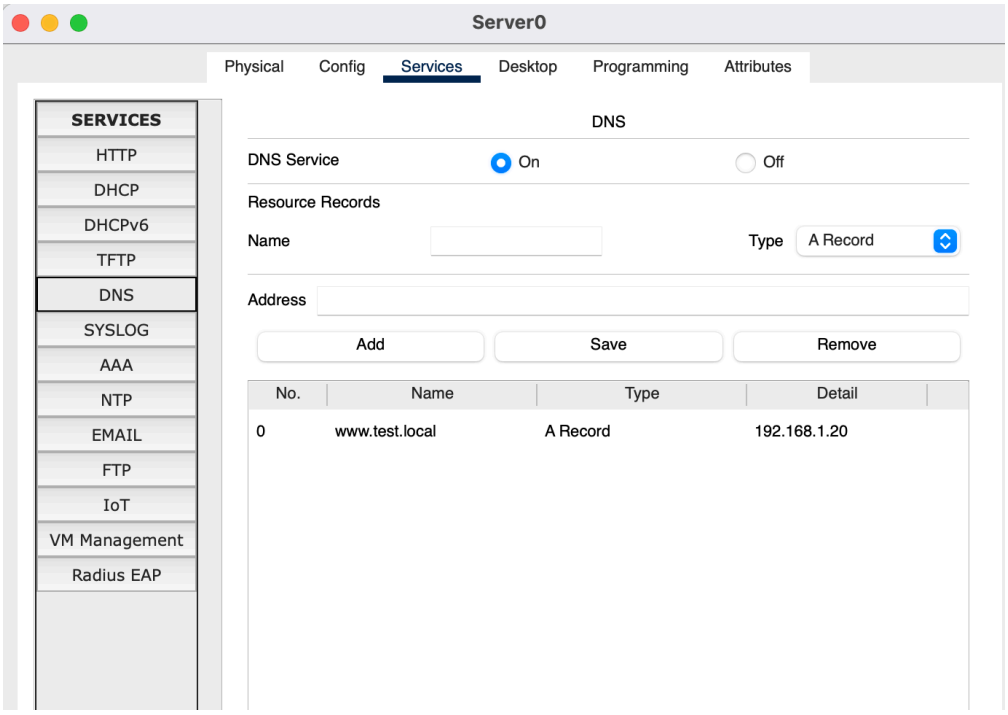


Figure 5 – HTTP Service Working

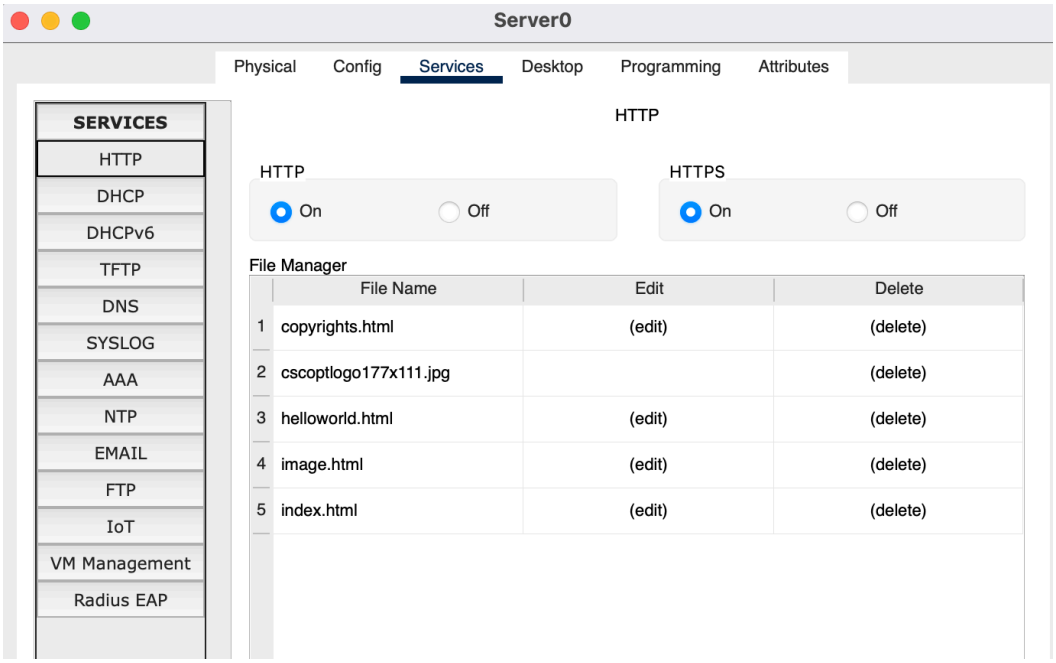


Figure 6 – FTP Service Working

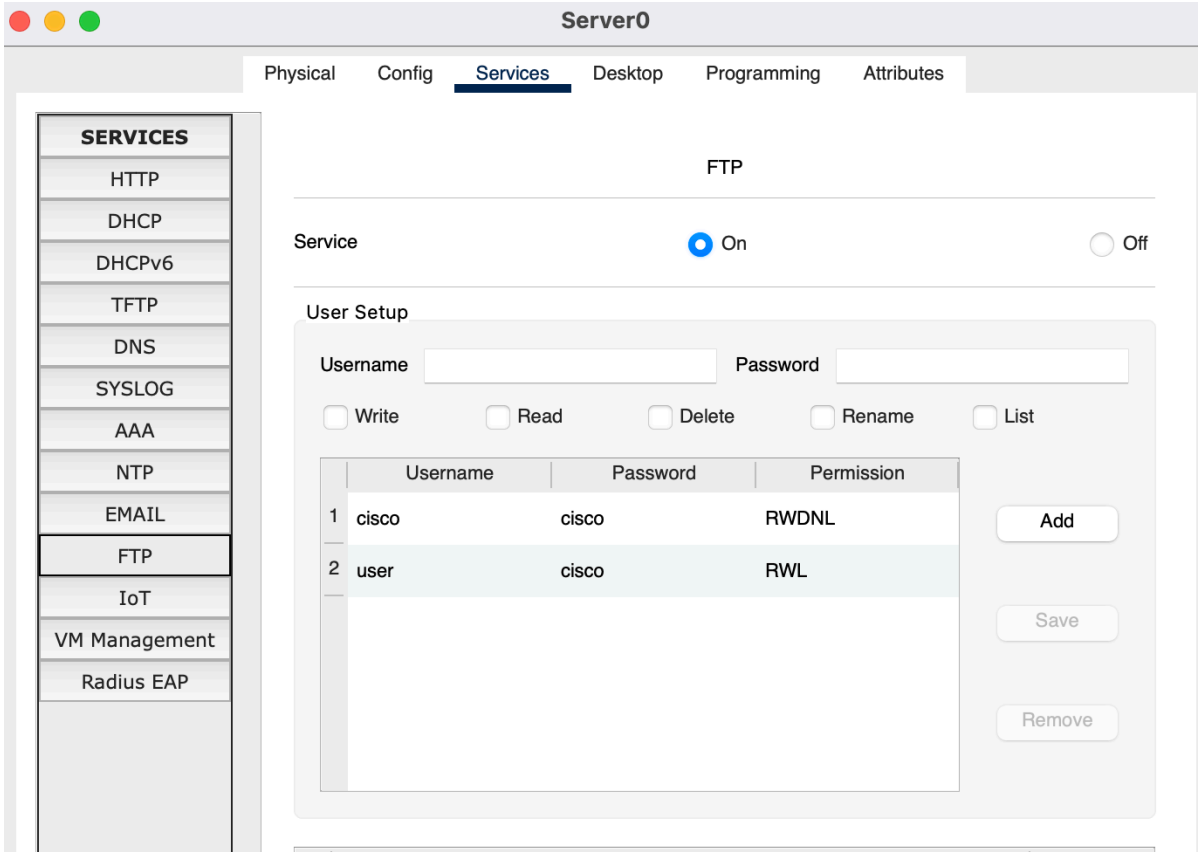


Figure 7 – Web Page Accessible

Figure 8 – FTP Directory

```
C:\>ftp www.test.local
Trying to connect...www.test.local
Connected to www.test.local
220- Welcome to PT Ftp server
Username:user
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from www.test.local:
0   : asa842-k8.bin                                    5571584
1   : asa923-k8.bin                                    30468096
2   : c1841-advipservicesk9-mz.124-15.T1.bin           33591768
3   : c1841-ipbase-mz.123-14.T7.bin                    13832032
4   : c1841-ipbasek9-mz.124-12.bin                     16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin           33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin           33591768
7   : c2600-i-mz.122-28.bin                            5571584
8   : c2600-ipbasek9-mz.124-8.bin                      13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin         50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin          33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin                  5571584
12  : c2800nm-ipbasek9-mz.124-8.bin                    15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin           33591768
14  : c2950-i6q412-mz.121-22.EA4.bin                   3058048
15  : c2950-i6q412-mz.121-22.EA8.bin                   3117390
16  : c2960-lanbase-mz.122-25.FX.bin                   4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin                 4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin                 4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin          8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin           10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin             33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin            83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin           505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG              159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG              184530138
26  : ir800-universalk9-bundle.SPA.156-3.M.bin         160968869
27  : ir800-universalk9-mz.SPA.155-3.M                 61750062
28  : ir800-universalk9-mz.SPA.156-3.M                 63753767
29  : ir800_yocto-1.7.2.tar                            2877440
30  : ir800_yocto-1.7.2_python-2.7.3.tar               6912000
31  : pt1000-i-mz.122-28.bin                           5571584
32  : pt3000-i6q412-mz.121-22.EA4.bin                  3117390
ftp>
```

Figure 9 – Ping Connectivity Test

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.test.local

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 2ms

C:\>
```

5. FTP Functionality Tests

FTP service tests were performed to verify authentication and file transfer.

Command	Function	Result
ftp	Connect to server	Successful
dir	List files	Successful
get	Download file	Successful
put	Upload file	Successful

These tests confirmed that the FTP service was working correctly before introducing the failure.

Figure 10 – Successful FTP Login

```
C:\>ftp www.test.local
Trying to connect...www.test.local
Connected to www.test.local
220- Welcome to PT Ftp server
Username:user
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

Figure 11 – Successful dir Command

```
ftp>dir

Listing /ftp directory from www.test.local:
0   : asa842-k8.bin                                    5571584
1   : asa923-k8.bin                                    30468096
2   : c1841-advipservicesk9-mz.124-15.T1.bin           33591768
3   : c1841-ipbase-mz.123-14.T7.bin                    13832032
4   : c1841-ipbasek9-mz.124-12.bin                     16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin           33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin           33591768
7   : c2600-i-mz.122-28.bin                            5571584
8   : c2600-ipbasek9-mz.124-8.bin                      13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin         50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin          33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin                  5571584
12  : c2800nm-ipbasek9-mz.124-8.bin                    15522644
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19  : c3560-advipservicesk9-mz.122-37.SE1.bin          8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin           10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin             33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin            83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin           505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG              159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG              184530138
26  : ir800-universalk9-bundle.SPA.156-3.M.bin         160968869
27  : ir800-universalk9-mz.SPA.155-3.M                 61750062
28  : ir800-universalk9-mz.SPA.156-3.M                 63753767
29  : ir800_yocto-1.7.2.tar                            2877440
30  : ir800_yocto-1.7.2_python-2.7.3.tar               6912000
31  : pt1000-i-mz.122-28.bin                           5571584
32  : pt3000-i6q4l2-mz.121-22.EA4.bin                  3117390
33  : test.txt                                         22
ftp>
```

Figure 12 – Successful get Command

```
ftp>get test.txt

Reading file test.txt from www.test.local:
File transfer in progress...

[Transfer complete - 22 bytes]

22 bytes copied in 0 secs
ftp>
```

Figure 13 – Successful put Command

```
ftp>put test.txt

Writing file test.txt to www.test.local:
File transfer in progress...

[Transfer complete - 22 bytes]

22 bytes copied in 0.078 secs (282 bytes/sec)
ftp>
```

6. Scenario 1 – DNS Failure

6.1 Problem Description

The PC could not access the server using the domain name **www.test.local**.

6.2 Diagnosis

Connectivity tests were performed:

- Ping to the server IP address worked correctly.
- Ping to the domain name failed.

This indicated that the problem was related to DNS name resolution.

Figure 14 – Incorrect DNS Configuration

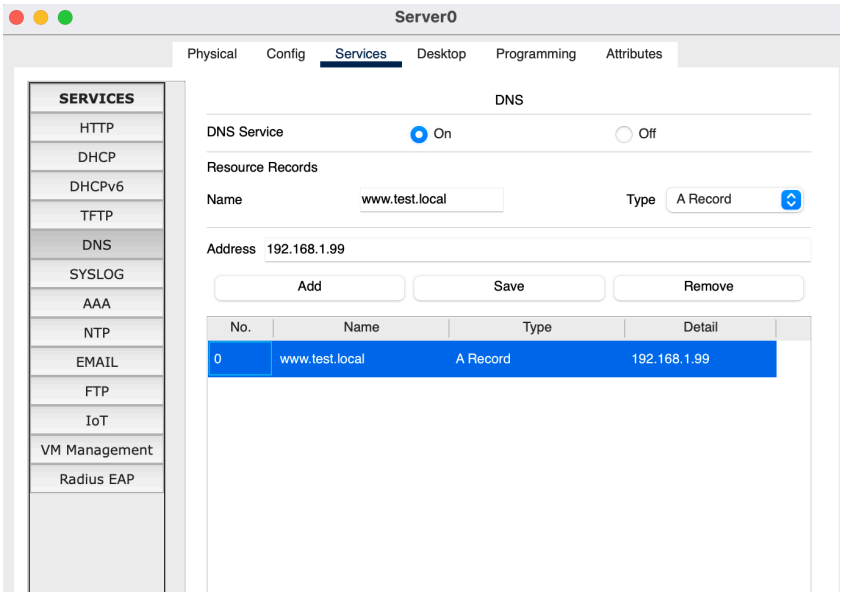


Figure 15 – Ping to Domain Failing

```
C:\>ping www.test.local

Pinging 192.168.1.99 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.99:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 16 – Ping to IP Working

```
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

6.3 Cause of the Problem

The DNS server IP address was incorrectly configured on the PC.

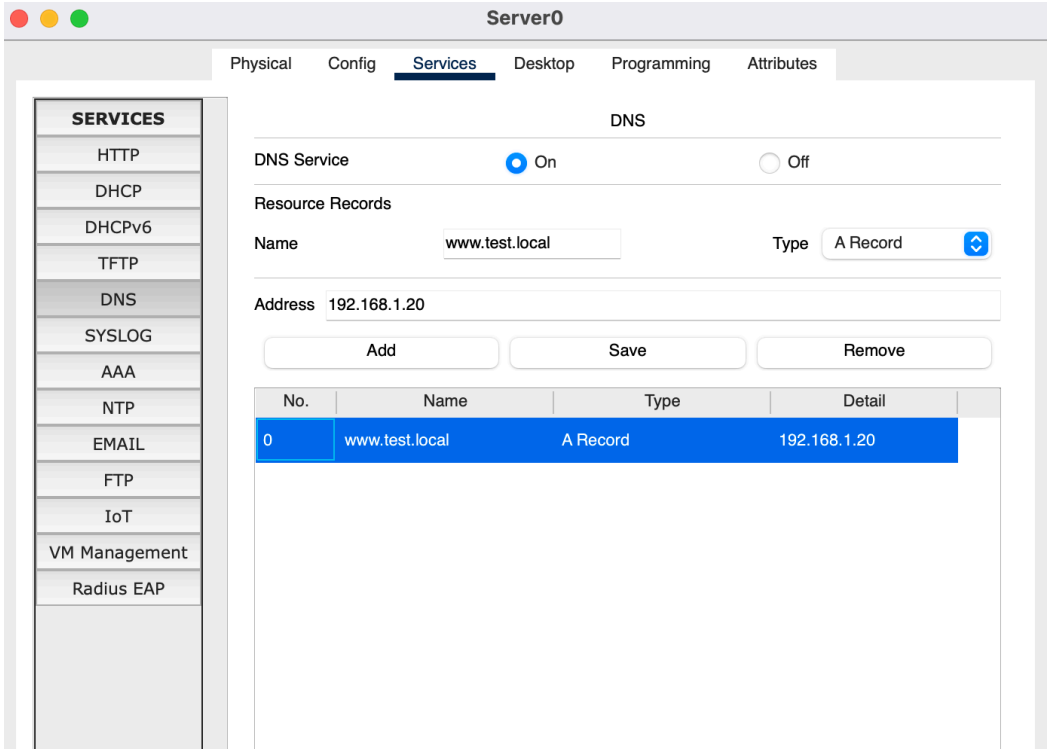
6.4 Solution

The DNS server IP address was corrected in the PC configuration.

Figure 17 – DNS Fixed

```
C:\>ping www.test.local

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 18 – Domain Access Working



6.5 Simulation Mode Analysis

In Simulation Mode, the DNS resolution process and the communication between the client and the server could be observed.

Figure 19 – DNS Simulation Failing

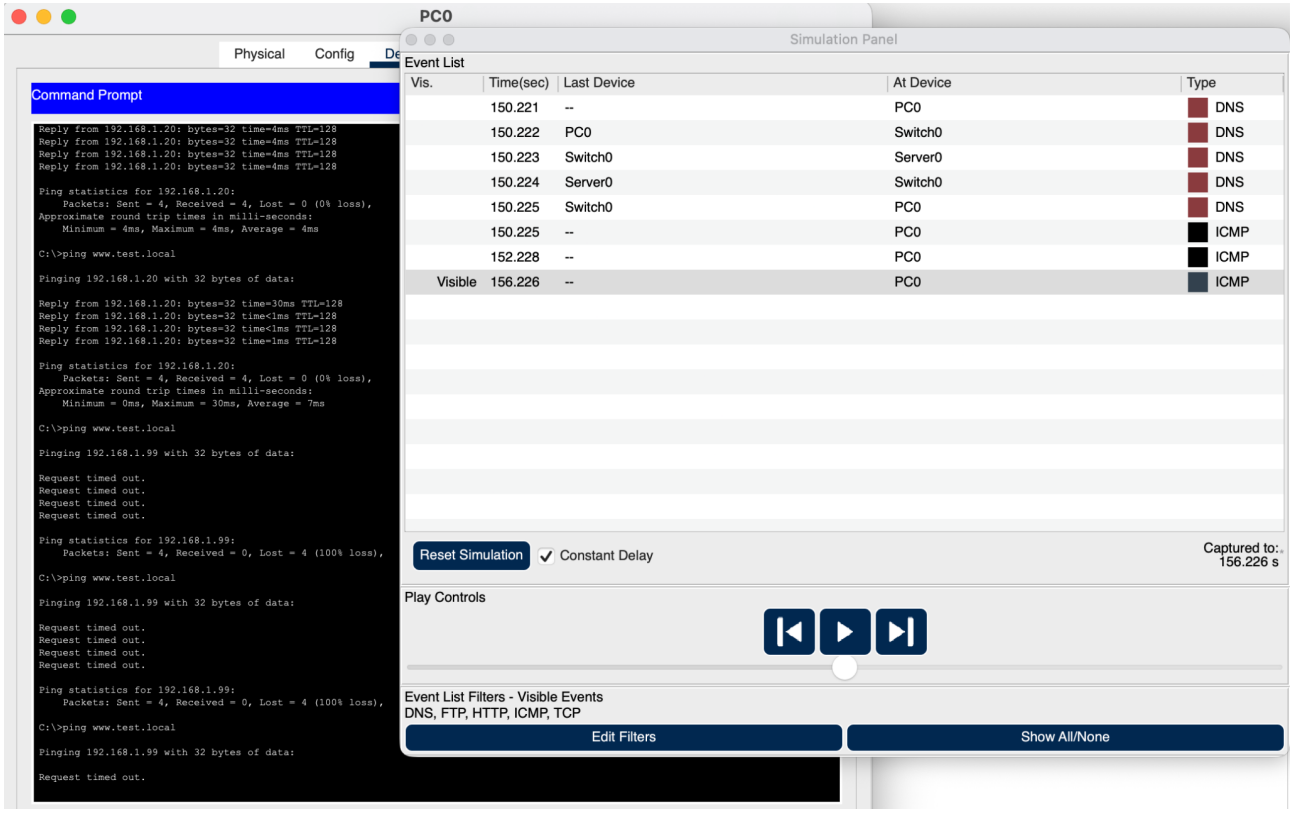
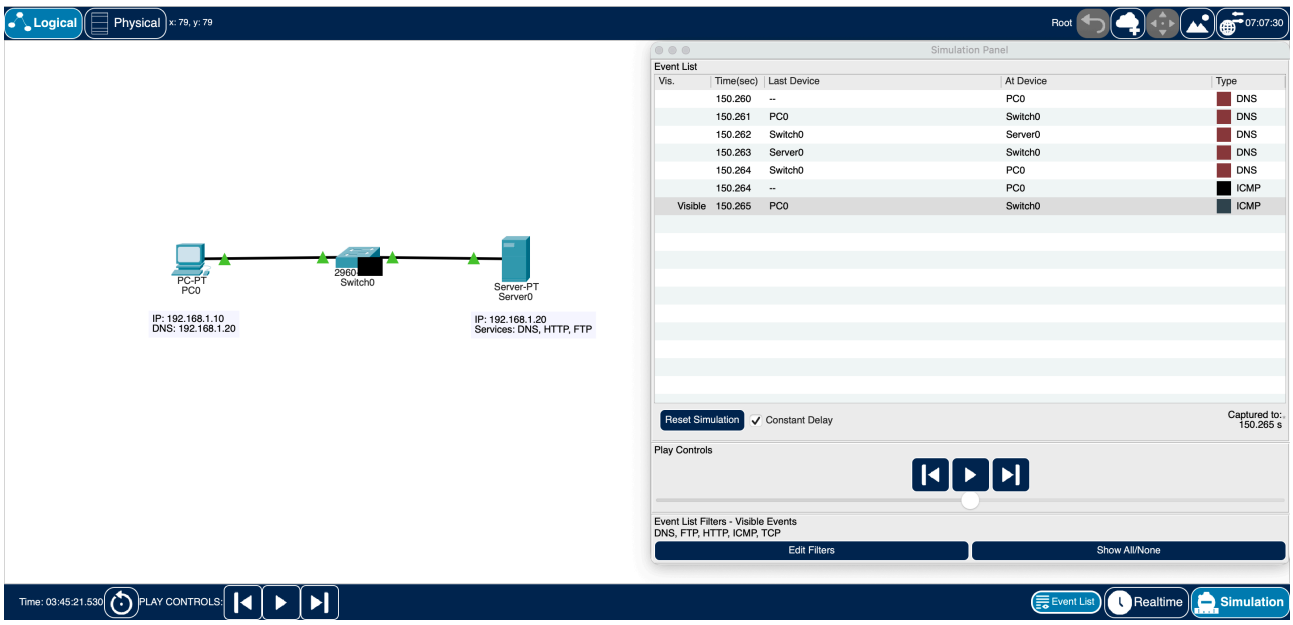


Figure 20 – DNS Simulation Working



7. Scenario 2 – HTTP Failure

7.1 Problem Description

The PC was able to ping the server, but could not access the web page.

7.2 Diagnosis

The message displayed in the browser indicated a connection error with the web server.

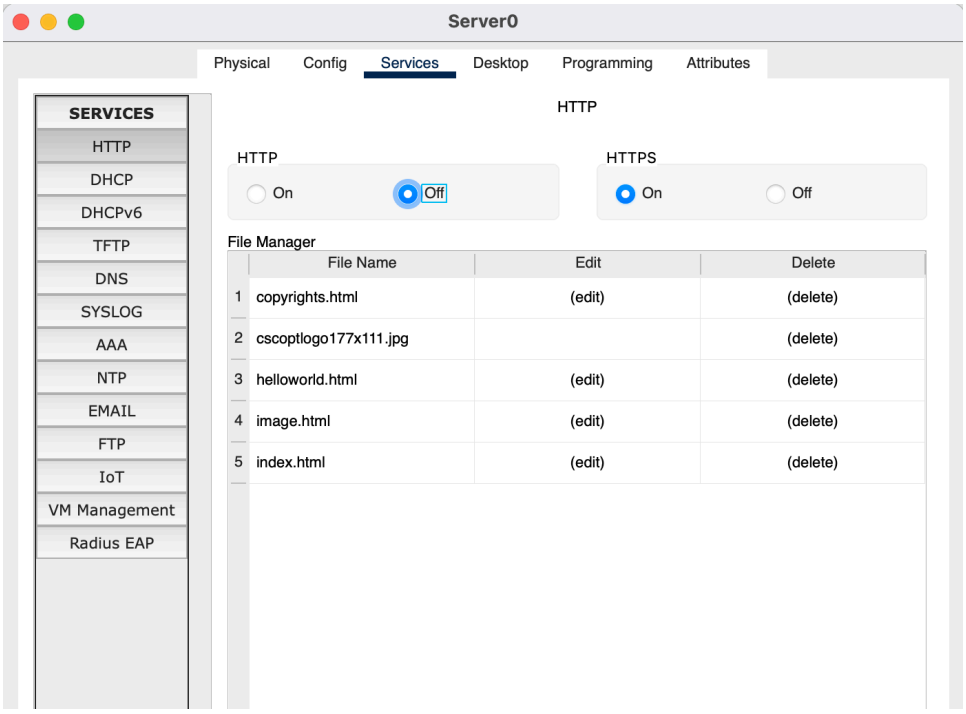
Figure 21 – Ping Works but HTTP Fails

```
C:\>ping www.test.local

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

7.3 Cause of the Problem

The HTTP service was disabled on the server.

Figure 22 – HTTP Service Disabled



7.4 Solution

The HTTP service was re-enabled on the server.

Figure 23 – Server Reset

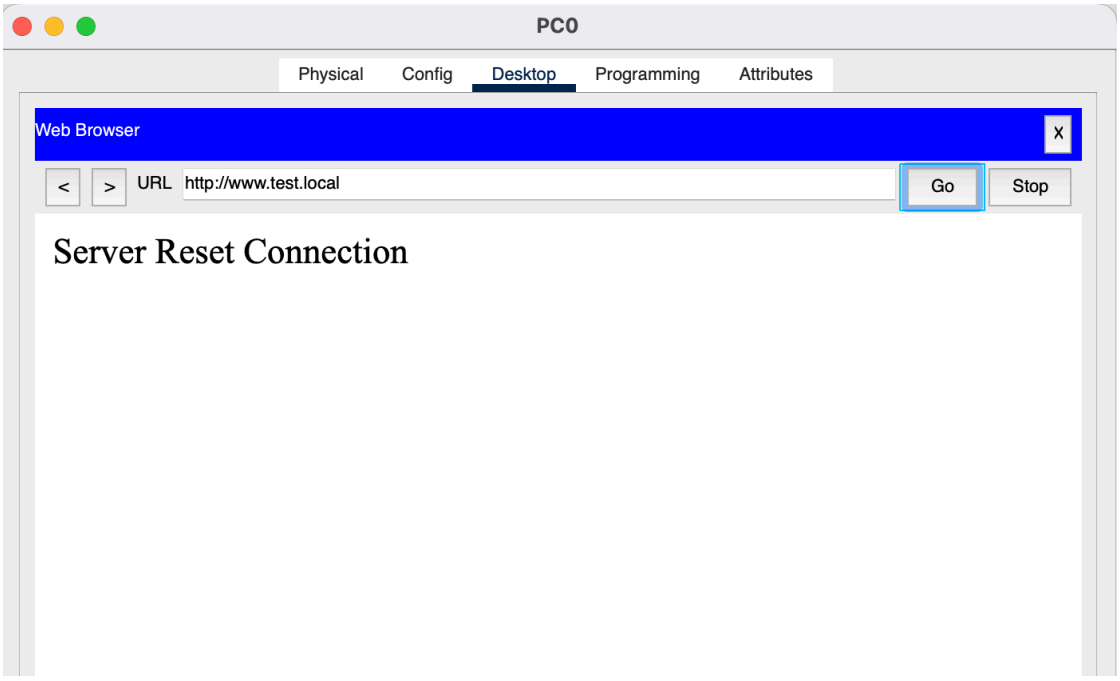


Figure 24 – HTTP Service Enabled

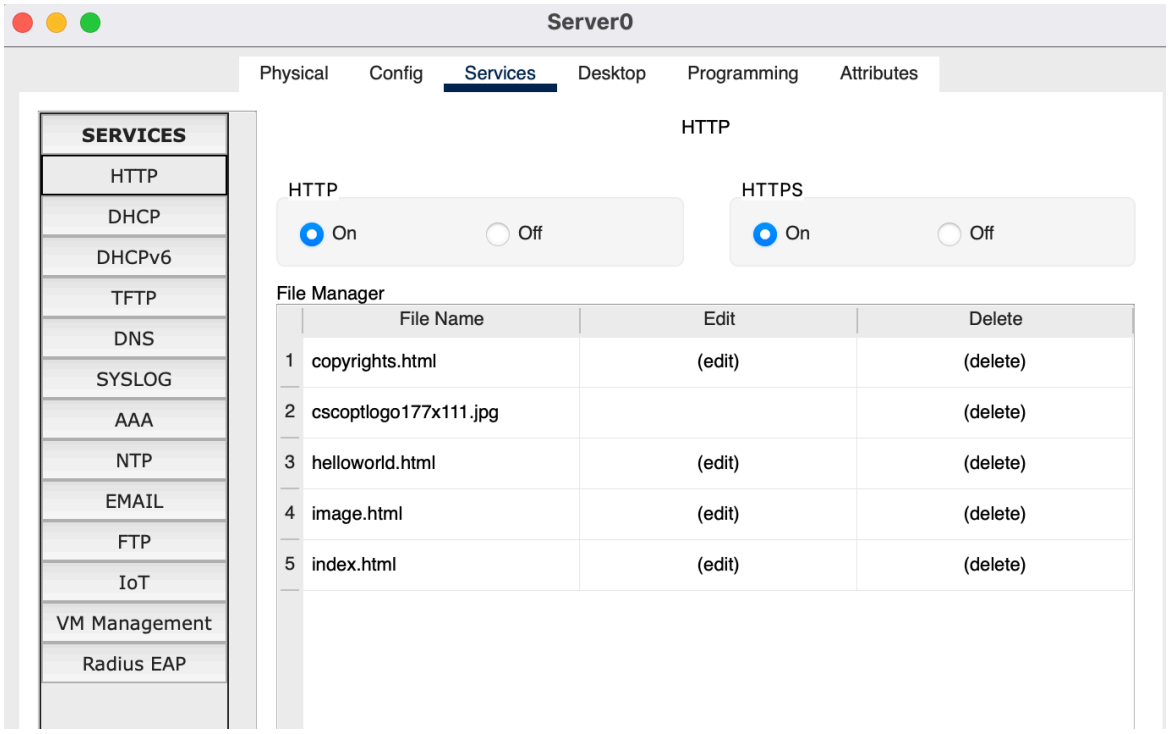
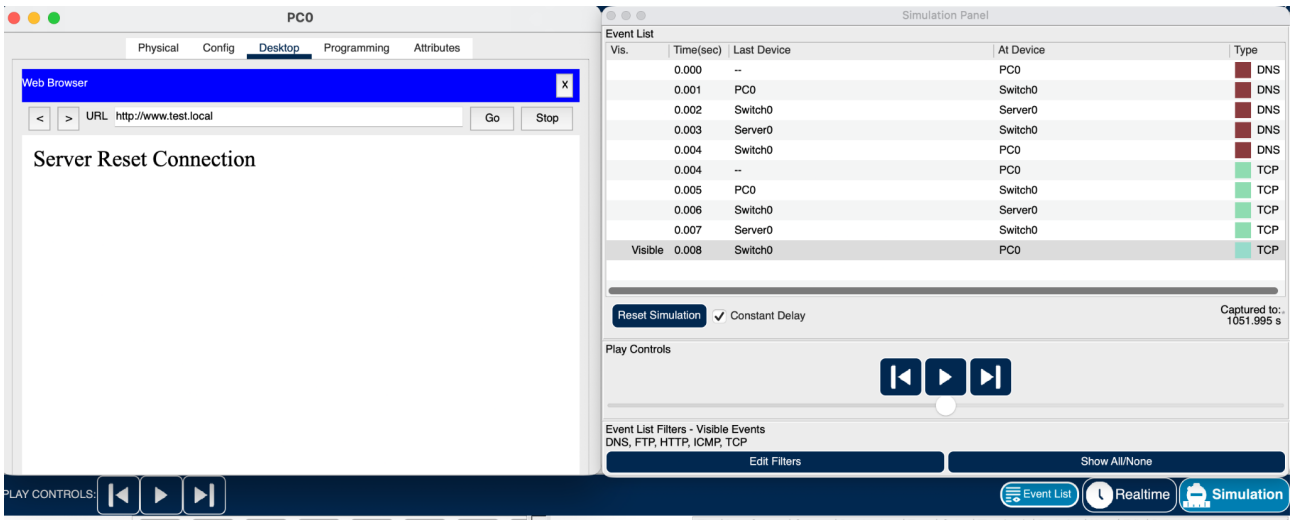


Figure 25 – Web Page Working

7.5 Simulation Mode Analysis

In Simulation Mode, TCP communication between the client and the server and the HTTP access attempt were observed.

Figure 26 – HTTP Simulation Failing



8. Scenario 3 – FTP Failure

8.1 Problem Description

The user was able to log in to the FTP server but could not list the directory contents.

8.2 Diagnosis

The **dir** command returned the error **550 Permission denied**, indicating a permissions problem.

Figure 27 – Incorrect FTP Permissions

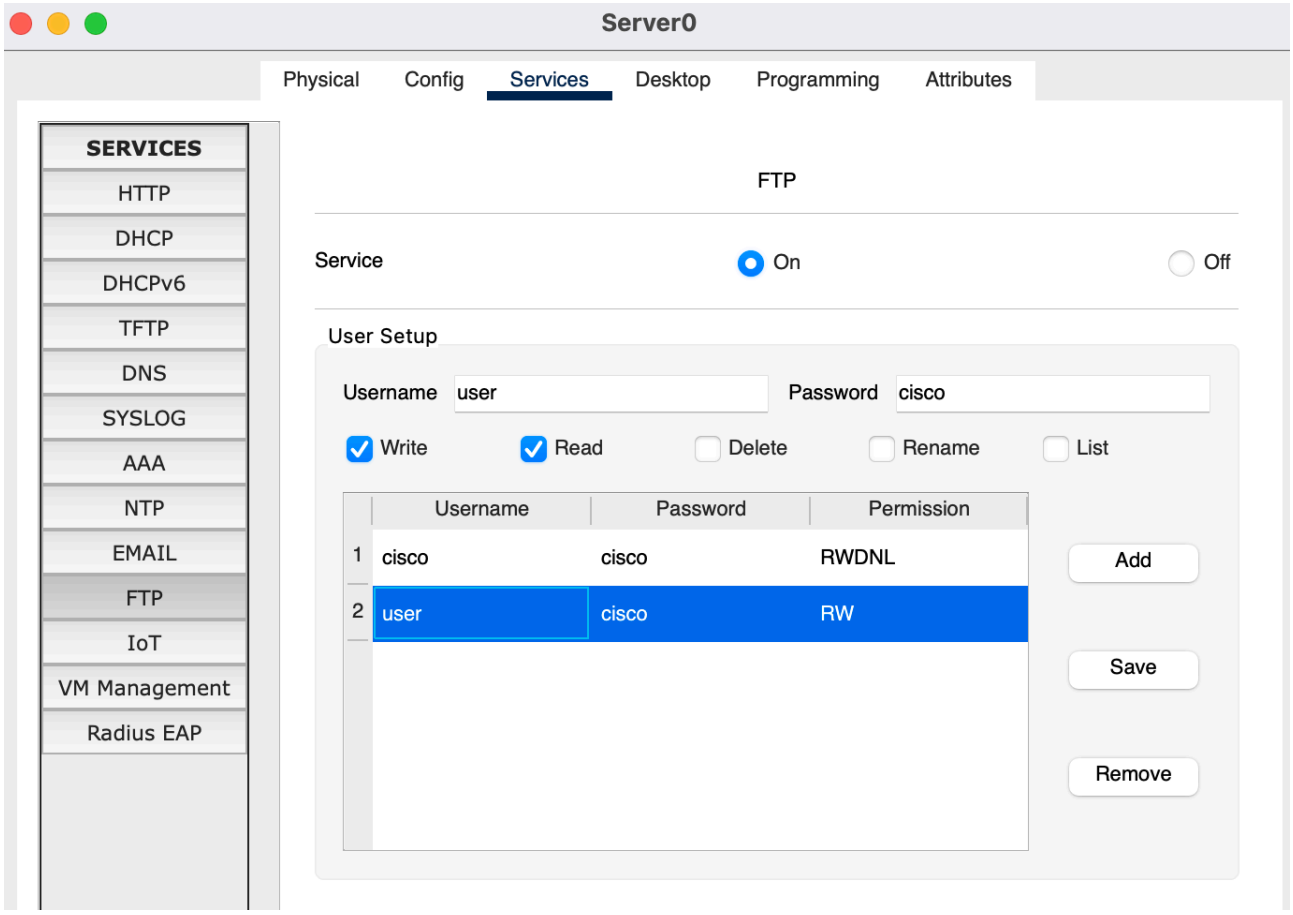


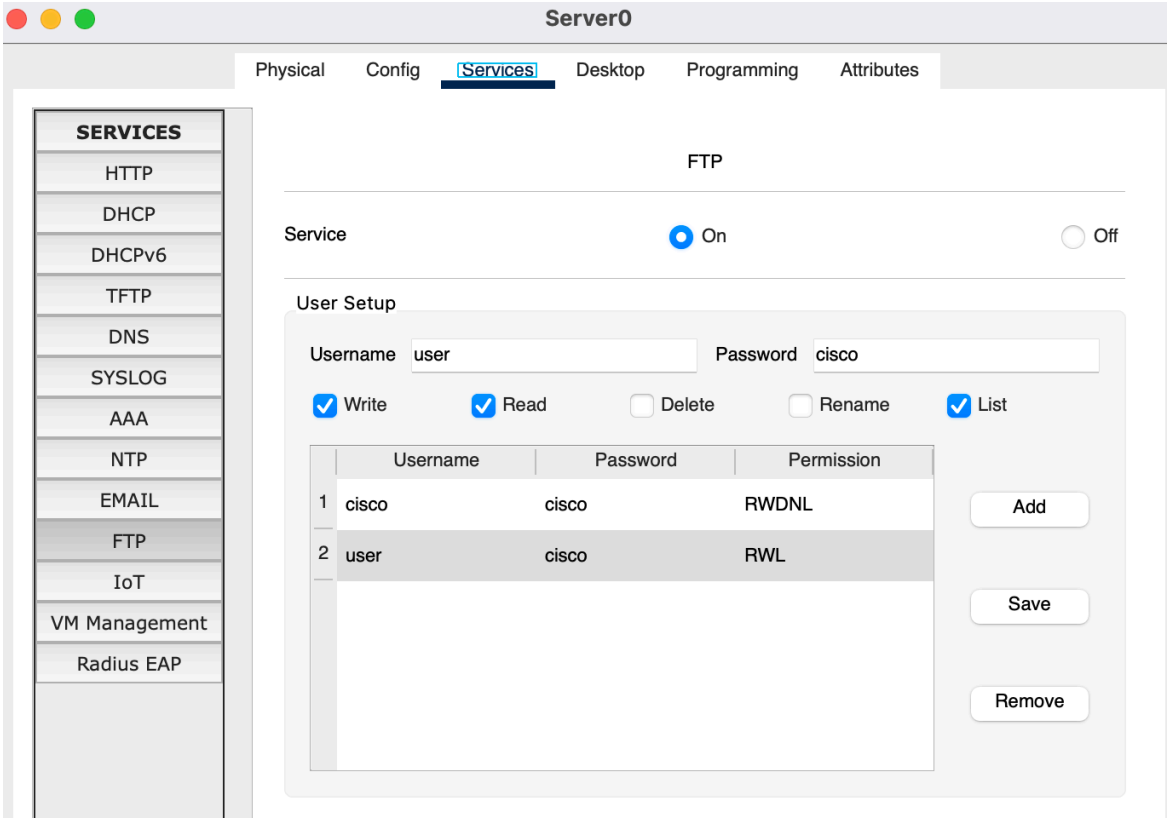
Figure 28 – Directory Listing Error

```
ftp>dir

Listing /ftp directory from www.test.local:
%Error ftp://www.test.local/ (No such file or directory Or Permission denied)
550-Requested action not taken. permission denied).

ftp>
```

8.3 Cause of the Problem

The FTP user LIST permission was disabled on the server.

Figure 29 – Permission Verification



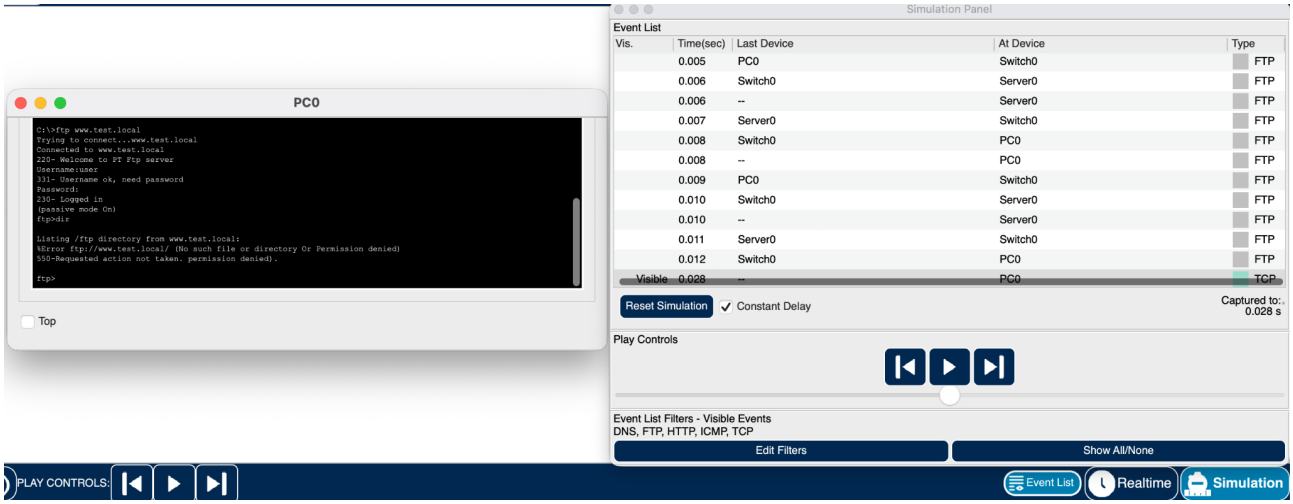
8.4 Solution

The LIST permission for the FTP user was re-enabled.

8.5 Simulation Mode Analysis

In Simulation Mode, the FTP LIST command and the server response indicating access denial were observed.

Figure 30 – FTP Simulation Failing



9. Conclusion

This laboratory made it possible to apply troubleshooting techniques to identify and solve failures in DNS, HTTP, and FTP services.

The following was observed:

- An error in DNS configuration prevents name resolution.
- A disabled HTTP service prevents access to the web page even if network connectivity exists.
- An FTP permissions problem allows authentication but prevents access to files.

The use of Simulation Mode in Cisco Packet Tracer allowed analysis of network traffic, identification of the protocols used, and understanding of how to detect failures in each service.

This laboratory helped develop network troubleshooting skills and highlighted the importance of verifying configuration, services, and permissions during the troubleshooting process.